

Perspective: Consistent resummation schemes for the one-particle Green's function

Johannes Tölle^{*,1,a)}, Arno Förster^{2,b)} and Pierre-François Loos^{*3,c)}

¹⁾Department of Chemistry, University of Hamburg, 22761 Hamburg, Germany; The Hamburg Centre for Ultrafast Imaging (CUI), Hamburg 22761, Germany

²⁾Department of Chemistry and Pharmaceutical Sciences, Vrije Universiteit, De Boelelaan 1105, 1081 HV Amsterdam, The Netherlands

³⁾Laboratoire de Chimie et Physique Quantiques (UMR 5626), Université de Toulouse, CNRS, Toulouse, France

Abstract: abstract abstract ...

I. INTRODUCTION

Taking a bottom-up perspective on electronic structure theory, the electron represents the fundamental building block whose behavior one aims to understand. By nature, electrons are indistinguishable and obey the Pauli exclusion principle, which fundamentally shapes their collective behavior in multi-electronic systems. Nevertheless, understanding the collective behavior of a system through its changes in the presence of an additional electron or hole is useful conceptually. Precisely, from this behavior, information about the ground and excited states of a system can be inferred.

Imagine injecting a hole in a system; depending on how strongly the electrons interact with this hole, one can deduce information of the electronic correlations present in the system. If the response is weak, the hole moves relatively freely through the system, essentially behaving as an independent particle. Conversely, if the response is strong, the hole becomes entangled with the surrounding electrons, which ultimately could result in a loss of its single-particle character.

The natural language that enables the description of this behavior is provided by the one-particle Green's function.

II. ONE-PARTICLE GREEN'S FUNCTION

As a sum-over-state expression for the interacting one-particle Green's function reads

$$G_{pq}(\omega) = \sum_N \langle \Psi_0 | a_p | \Psi_N \rangle (\omega - \Delta E_N^+)^{-1} \langle \Psi_N | a_q^\dagger | \Psi_0 \rangle + \sum_M \langle \Psi_0 | a_q^\dagger | \Psi_M \rangle (\omega - \Delta E_M^-)^{-1} \langle \Psi_M | a_p | \Psi_0 \rangle \quad (1)$$

with $a_q^\dagger$ and a_p denoting fermionic creation and annihilation operators, and $\Delta E_N^+ = E_N^+ - E_0$ and $\Delta E_M^- = E_0 - E_M^-$. With + and - referring to the energy of system with an added/removed electron.

As an alternative to Eq. (1), the one-particle Green's function can be determined through the Dyson equation

$$G_{pq}(\omega) = G_{pq}^0(\omega) + \sum_{rs} G_{pr}^0(\omega) \Sigma_{rs}(\omega) G_{sq}(\omega), \quad (2)$$

with $G_{pq}^0(\omega)$ denoting the non-interacting Green's function

$$G_{pq}^0(\omega) = \delta_{pq} \frac{1 - n_p}{\omega + \epsilon_p} + \delta_{pq} \frac{n_p}{\omega - \epsilon_p}, \quad (3)$$

with n_p denoting the occupation number from the Fermi-Dirac distribution. We assume the zero temperature limit throughout, n_p associated with a mean-field orbital p is therefore either 0 (for an unoccupied orbital) or 1 (for an occupied orbital). Eq. (2) can also be rearranged to (in explicit matrix form)

$$\mathbf{G}(\omega) = \frac{1}{\mathbf{G}_0^{-1}(\omega) - \mathbf{\Sigma}(\omega)}. \quad (4)$$

The connection between the sum-over-state expression and the Dyson equation is made explicit in the following. For this, we define a new vacuum $|\Psi_0\rangle$

$$|\Psi_0\rangle = e^\sigma |0\rangle, \quad (5)$$

via exponential parametrization of the reference mean-field state $|0\rangle$. σ denotes an anti-Hermitian particle-hole excitation operator

$$\sigma = \hat{T} - \hat{T}^\dagger, \quad (6)$$

with

$$\hat{T} = \sum_v t_v \bar{\tau}_v. \quad (7)$$

$\bar{\tau}_v$ represents an arbitrary particle-hole excitation operator ($\bar{a}_a^\dagger \bar{a}_i + \dots$). Note that the amplitudes t_v [Eq. (7)] are determined by projection and satisfy the general Brioullin conditions

$$\langle \Psi_0 | [\hat{H}, \bar{\tau}_v - \bar{\tau}_v^\dagger] | \Psi_0 \rangle = 0. \quad (8)$$

In the following, occupied orbitals are referred to with subscripts $i, j, k, \dots$, virtual orbitals with $a, b, c, \dots$ and arbitrary orbitals with $p, q, r, \dots$. Furthermore, the particle and hole creation and annihilation operators acting on the vacuum are (still obeying their anti-commutation relations)

$$\bar{a}_p = e^\sigma a_p e^{-\sigma} \quad (9)$$

$$\bar{a}_p^\dagger = e^\sigma a_p^\dagger e^{-\sigma}. \quad (10)$$

Within this formalism an arbitrary excited state N is obtained as

$$|\Psi_N\rangle = \hat{R}_N |\Psi_0\rangle. \quad (11)$$

^{a)}Electronic mail: johannes.toelle@uni-hamburg.de

^{b)}Electronic mail: a.t.l.foerster@vu.nl

^{c)}Electronic mail: loos@irsamc.ups-tlse.fr

$\hat{R}_N$ is given as

$$\begin{aligned}\hat{R}_N |\Psi_0\rangle &= \sum_{\nu} R_{\nu}^N \tilde{r}_{\nu} |\Psi_0\rangle \\ &= \sum_{\nu} R_{\nu}^N e^{\sigma} \hat{r}_{\nu} e^{-\sigma} |\Psi_0\rangle = \sum_{\nu} R_{\nu}^N e^{\sigma} |\Phi_{\nu}\rangle,\end{aligned}\quad (12)$$

with R_{ν}^N denoting the expansion coefficients so that

$$|\Psi_N\rangle = \sum_{\nu} R_{\nu}^N e^{\sigma} \hat{r}_{\nu} |0\rangle. \quad (13)$$

Note that $\hat{r}_{\nu}$ is chosen according to the excited state of interest, e.g., for charged ($N \pm 1$) excitations

$$\hat{r}_{\nu}^{-} = \{a_i, a_b^{\dagger} a_j a_i, \dots\} \quad (14)$$

$$\hat{r}_{\nu}^{+} = \{a_a^{\dagger}, a_a^{\dagger} a_b^{\dagger} a_j, \dots\} \quad (15)$$

$$\hat{r}_{\nu} = \{\hat{r}_{\nu}^{-}\} \cup \{\hat{r}_{\nu}^{+, \dagger}\} \quad (16)$$

so that

$$\begin{aligned}1 &= \sum_{\nu} \tilde{r}_{\nu} |\Psi_0\rangle \langle \Psi_0| \tilde{r}_{\nu}^{\dagger} \\ 1 &= \sum_{\nu} e^{\sigma} \hat{r}_{\nu} |0\rangle \langle 0| \hat{r}_{\nu}^{\dagger} e^{-\sigma}\end{aligned}\quad (17)$$

is satisfied. The Hamiltonian for that new vacuum is defined in terms of the new set of creation and annihilation operators

$$\begin{aligned}\tilde{H} &= e^{\sigma} \hat{H} e^{-\sigma} \\ &= \sum_{pq} F_{pq} \tilde{a}_p^{\dagger} \tilde{a}_q + \frac{1}{4} \sum_{pqrs} V_{pqsr} \tilde{a}_p^{\dagger} \tilde{a}_q^{\dagger} \tilde{a}_r \tilde{a}_s.\end{aligned}\quad (18)$$

Within the newly defined quantities excitation energy differences ΔE_N are evaluated using the EOM approach

$$\begin{aligned}\langle \Psi_0 | [\tilde{r}_{\nu}^{\dagger}, [\tilde{H}, \sum_{\mu} R_{\mu}^N \tilde{r}_{\mu}]] | \Psi_0 \rangle \\ = \Delta E_N \langle \Psi_0 | [\tilde{r}_{\nu}^{\dagger}, \sum_{\mu} R_{\mu}^N \tilde{r}_{\mu}] | \Psi_0 \rangle,\end{aligned}\quad (19)$$

which can be rewritten to

$$\begin{aligned}\langle 0 | [\hat{r}_{\nu}^{\dagger}, [\hat{H}, \sum_{\mu} R_{\mu}^N \hat{r}_{\mu}]] | 0 \rangle \\ = \langle 0 | [\hat{r}_{\nu}^{\dagger}, \sum_{\mu} R_{\mu}^N \hat{r}_{\mu}] | 0 \rangle \Delta E_N.\end{aligned}\quad (20)$$

Eq. (20) resembles the common EOM configuration interaction approach and can be explicitly written in matrix form

$$\mathbf{H}\mathbf{R} = \mathbf{S}\mathbf{R}\mathbf{E} \quad (21)$$

$$\Leftrightarrow \tilde{\mathbf{H}}\mathbf{R} = \mathbf{R}\mathbf{E} \quad (22)$$

(assuming $\mathbf{S}$ is invertible and a symmetric transformation).

With the above definitions, Eq. (1) reads

$$\begin{aligned}G_{pq}(\omega) &= \sum_N \langle \Psi_0 | \tilde{a}_p | \Psi_N \rangle (\omega - \Delta E_N^+)^{-1} \langle \Psi_N | \tilde{a}_q^{\dagger} | \Psi_0 \rangle \\ &+ \sum_M \langle \Psi_0 | \tilde{a}_q^{\dagger} | \Psi_N \rangle (\omega - \Delta E_M^-)^{-1} \langle \Psi_N | \tilde{a}_p | \Psi_0 \rangle,\end{aligned}\quad (23)$$

which can be rearranged to

$$\mathbf{G}_{11}(\omega) = \begin{pmatrix} \mathbf{1}_{11} & \mathbf{0} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_{11} \\ \mathbf{0} \end{pmatrix}, \quad (24)$$

where the subscript 11 denotes the p/h subspace and 2 refers to all higher order excitations, i.e., hhp, pph etc.. The connection to the Dyson equation becomes apparent by rewriting Eq. (24)

$$\begin{aligned}\mathbf{G}_{11}(\omega) &= [\omega \mathbf{1}_{11} - \mathbf{H}_{11} - \mathbf{H}_{12}(\omega \mathbf{1}_{22} - \mathbf{H}_{22})^{-1} \mathbf{H}_{21}]^{-1} \\ \Rightarrow \mathbf{G}(\omega) &= [\omega - \mathbf{F} - \mathbf{\Sigma}(\infty) - \tilde{\mathbf{\Sigma}}(\omega)]^{-1} \\ \Leftrightarrow \mathbf{G}(\omega) &= [\omega - \mathbf{F} - \mathbf{\Sigma}(\omega)]^{-1} \\ \Leftrightarrow \mathbf{G}(\omega) &= [\mathbf{G}_0^{-1}(\omega) - \mathbf{\Sigma}(\omega)]^{-1}.\end{aligned}\quad (25)$$

Consequently, we have demonstrated the connection of the effective Hamiltonian used for the determination of single-particle excitation energies and the Dyson equation of the one-particle Green's function. For generality, we have partitioned H_{11} into the bare Fock matrix and additional screening effects summarized in $\Sigma(\infty)$ (for example HF exchange is often considered to be part of $\Sigma(\infty)$). Furthermore, particle and hole space are now intrinsically coupled in $\Sigma(\omega)$.

The route described above is not the strategy commonly used in quantum chemistry for the determination of the one-particle Green's function. Instead, the original definition of creation and annihilation operators is kept, so that the EOM reads

$$\begin{aligned}\langle 0 | [\hat{r}_{\nu}^{\dagger}, [\tilde{H}, \sum_{\mu} R_{\mu}^N \hat{r}_{\mu}]] | 0 \rangle \\ = \langle 0 | [\hat{r}_{\nu}^{\dagger}, \sum_{\mu} R_{\mu}^N \hat{r}_{\mu}] | 0 \rangle \Delta E_N,\end{aligned}\quad (26)$$

with $\tilde{H}$ denoting the unitary transformed Hamiltonian

$$\tilde{H} = e^{-\sigma} \hat{H} e^{\sigma} \quad (27)$$

and the one-particle Green's function written as

$$\begin{aligned}G_{pq}(\omega) &= \sum_{\nu\mu} \langle 0 | e^{-\sigma} a_p e^{\sigma} | \Phi_{\nu} \rangle [(\omega - \mathbf{H}^+)^{-1}]_{\nu\mu} \langle \Phi_{\mu} | e^{-\sigma} a_q^{\dagger} e^{\sigma} | \Psi_0 \rangle \\ &+ \sum_{\nu\mu} \langle 0 | e^{-\sigma} a_q^{\dagger} e^{\sigma} | \Phi_{\nu} \rangle [(\omega - \mathbf{H}^-)^{-1}]_{\nu\mu} \langle \Phi_{\mu} | e^{-\sigma} a_p e^{\sigma} | \Psi_0 \rangle,\end{aligned}\quad (28)$$

with $\mathbf{H}^+$, $\mathbf{H}^-$ denoting the EOM Hamiltonian in the complete $N \pm 1$ excitation basis. Note that no coupling between these two sectors occurs due to Eq. (8). Furthermore, Φ_{ν} denotes excited Slater determinants in the $N \pm 1$ sectors [compare Eq. (12)]. Next, we write Eq. (28) compactly as

$$\begin{aligned}\mathbf{G}_{11}(\omega) &= \\ &= \begin{pmatrix} \mathbf{M}_{11}^{\dagger} & \mathbf{M}_{21}^{\dagger} \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{M}_{11} \\ \mathbf{M}_{21} \end{pmatrix}\end{aligned}\quad (29)$$

(note that in the blocks the $N+1$ and $N-1$ are still decoupled). $\mathbf{M}_{11}$ denotes contributions from $\langle \Phi_{\mu} | e^{-\sigma} a_q^{\dagger} e^{\sigma} | \Psi_0 \rangle$ and

$\langle \Phi_\mu | e^{-\sigma} a_p e^\sigma | \Psi_0 \rangle$ in the 1h-1p sector and $\mathbf{M}_{12}$ in all higher excitation sectors. It becomes clear that the terms such as $\langle 0 | e^{-\sigma} a_p e^\sigma | \Phi_\nu \rangle$ in Eq. (28) prohibit a simple projection of the decoupled EOM Hamiltonians $\mathbf{H}^+/\mathbf{H}^-$ in the 1-h/1-p space as in Eq. (24). Consequently the precise connection to the Dyson equation can be only made approximately by separating $e^{-\sigma} a_q^\dagger e^\sigma$ into zeros order term a_q and higher order terms of the BCH. Consequently, Eq. (29) is rearranged to

$$\begin{aligned} \mathbf{G}_{11}(\omega) = & \begin{pmatrix} \mathbf{1}_{11} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_{11} \\ \mathbf{0} \end{pmatrix} \\ & + \begin{pmatrix} \tilde{\mathbf{M}}_{11}^\dagger & \mathbf{M}_{21}^\dagger \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \tilde{\mathbf{M}}_{11} \\ \mathbf{M}_{21} \end{pmatrix}. \quad (30) \end{aligned}$$

From the above considerations for the one-particle Green's function it becomes apparent that a ground-state p/h transformation of the Hamiltonian $\hat{H}$ prohibits the determination of $\mathbf{G}$ through the Dyson equation [Eq. (2)] exactly. Instead, the different components of $\mathbf{G}$ have to be determined separately. (See also the work by Tarantelli and Cederbaum¹).

III. HEDIN VS PARQUET

REFERENCES

¹A. Tarantelli and L. Cederbaum, "Configuration-interaction formulation of the dyson equation," Phys. Rev. A **45**, 2790 (1992).